Characterization of the Immersion Factor for a Series of In-Water Optical Radiometers

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ABSTRACT

Spectral immersion factors, $I_f(\lambda)$, account for the difference between the in-air and in-water absolute response of submersible radiometers and are required to properly apply the in-air absolute calibration of the sensor when used underwater. The use of the so-called typical values for a series of in-water radiometers is a source of uncertainty because of intrinsic instrument-to-instrument differences in the optics. To investigate this source of uncertainty, in addition to uncertainties associated with determining the immersion factors using a laboratory method, a sample of nine radiometers from the same series of instruments was characterized by three different laboratories. A comparison of the immersion factor characterizations from the three laboratories indicates average intralaboratory measurement repeatabilities ranging from 0.3% to 0.6%, which were evaluated using multiple characterizations of the same reference radiometer and defined by two standard deviations. Interlaboratory relative uncertainties, evaluated with $I_f(\lambda)$ data from the nine sample radiometers, show average percent differences ranging from −0.5% to 0.6%. The dispersion of $I_f(\lambda)$ values across all the radiometers show values up to 5% in the red part of the spectrum with a spectral average of 2% (defined by two standard deviations). Typical $I_f(\lambda)$ values, computed with data from the so-called trusted radiometers (i.e., those not showing extreme outlier values), are also presented with their maximum uncertainties and a discussion on their spectral dependence.

1. Introduction

The absolute response of an optical instrument is different when it is used in air or in water, because of the refractive indices of the two media. In the case of an in-water radiometer designed for radiance measurements, the change in response characterizing in-air and in-water measurements is mostly due to (i) the relative difference in transmittance through the glass window located in front of the aperture and (ii) the relative change in the solid angle field of view. In the case of an in-water radiometer designed for irradiance measurements (using collectors made of a diffusing material and exhibiting a cosine angular response), the change in absolute response is primarily caused by the relative changes in the reflection properties of the air-collector and water-collector interfaces, and specifically by (i) a relative change in the reflection coefficient at the external surface of the collector (i.e., the so-called external reflection factor) and (ii) a relative change in the reflection coefficient of the internal surface of the collector (i.e., the so-called internal reflection factor).

The differences in absolute response caused by immersion effects are accounted for through spectral multiplication coefficients—the so-called immersion factors—applied to the in-air absolute radiometric calibration coefficients. The immersion factors have a value of 1 for in-air measurements and are greater than 1 (as a function of the material and geometric features of the optics) for in-water measurements. Spectral immersion factors for radiometers measuring in-water radiance are usually computed from the refractive indices of water

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and the glass window. Spectral immersion factors for radiometers measuring in-water irradiance need to be experimentally determined, because of the complex function relating the refractive index of water and the optical and geometrical characteristics of the collector.

a. Background

Early studies on immersion effects were carried out by Atkins and Poole (1933). They made an attempt to describe the internal and external reflection factors for an opal glass diffuser. To experimentally estimate these reflection contributions, they used a gas-filled lamp as a light source to vertically illuminate the diffuser when it was dry and wet, that is, in-air and in-water covered with different depths of distilled water, respectively. The authors proposed a constant immersion factor of 1.09 for opal glass diffusers to compensate for instrument sensitivity loss when operated in the water with respect to in the air. More recently, Berger (1958) presented (i) a discussion of the immersion effects in the presence of a thin layer of water producing direct reflections between the external surface of the diffuser and the water subsurface (a limiting factor in the characterization of the immersion factor) and (ii) a description of a simple method for experimentally determining the immersion factor of disk-shaped diffusers of radius r_d based on a wide blackened funnel to hold pure water above the diffuser with a water depth of at least 0.9 r_d. Notable contributions from Berger (1958) included observations and theoretical data on the immersion effects for different disk-shaped diffusers made of silicate and plastic glass showing a wide range of variations for the immersion factor.

Data from Berger (1961) were later used by Westlake (1965) to extensively describe the reflection–refraction processes occurring at the air–diffuser and at the water–diffuser interfaces in the presence of thin or deep layers of water. Westlake (1965) presented estimates of the different internal and external reflection contributions and suggested a constant immersion factor of 1.19 for opal glass, significantly higher than that proposed earlier by Atkins and Poole (1933).

A comprehensive description of a protocol for the experimental characterization of the immersion factor of in-water irradiance collectors was given by Smith (1969). The protocol, which included vertical measurements in air and in water with different depths of water above the diffuser, suggested the use of a collimated beam as a light source to avoid changes in the energy falling on the collector when different water depths were used. Smith (1969) presented a spectral characterization of the immersion factor of a cosine collector made of clear Plexiglas bonded together with translucent Plexiglas (with the latter in contact with water). The author determined immersion factors ranging within 1.34–1.22, almost linearly varying in the spectral interval 400–750 nm (as summarized in Tyler and Smith 1970), respectively, and qualitatively explained the spectral values with a dependence on the absorbance of the collector.

Petzold and Austin (1988) revised the protocol for characterizing immersion factors for irradiance collectors. They proposed using a lamp as a light source by introducing a geometric correction factor originally applied by Aas (1969) that, as a function of the lamp-collector distance, water depth, and water refractive index, minimizes effects due to changes in the energy falling on the collector as a function of a change in water depth. Applying the former method, Mueller (1995) analyzed $I_f(\lambda)$ for collectors made of Plexiglas and Teflon for several radiometers from the same manufacturer. Mueller reported $I_f(\lambda)$ values almost linearly decreasing with wavelength and ranging, on the average, within 1.38–1.32 in the spectral interval 406–670 nm, respectively.

b. Rationale for the present work and objectives

The immersion factors for in-water optical radiometers are commonly assumed to be specific for each series of instruments and so-called typical spectral immersion factors are generally provided to users. The work of Mueller (1995) on the characterization of 12 irradiance collectors on a different series of marine radiometers showed a dispersion in spectral immersion factors of the same series of instruments ranging from 3% to 5% (defined by one standard deviation, i.e., 1σ) with values as large as 10%. The study, based on repeated measurements of the same collectors, showed uncertainties in measurement repeatability (defined by 1σ) of 1%. The major conclusion of the study was that accurate in-water radiometry requires an experimental characterization of each individual irradiance collector.

The possible nonapplicability of typical immersion factors was reported by Dierssen and Smith, (1996) for Profiling Reflectance Radiometers (PRRs, manufactured by Biosperical Instruments Inc.), and in Zibordi and Berthon (2001), for Ocean Color Irradiance 200 series radiometers (OCI-200, manufactured by Satlantic, Inc.). Uncertainties in the spectral immersion factors of in-water irradiance collectors affect the following:

1) absolute downward $E_d(z, \lambda)$ and upward $E_u(z, \lambda)$ irradiances at wavelength λ, and depth z;
2) $Q_n(z, \lambda)$ factors and the irradiance reflectance $R(z, \lambda)$ when computed with $E_u(z, \lambda)$ and $E_d(z, \lambda)$ obtained from different radiometers;
3) normalized water-leaving radiance $L_{WN}(\lambda)$ and remote sensing reflectance $R_{rs}(\lambda)$, when $E_d(0^+, \lambda)$ is computed from subsurface $E_d(0^-, \lambda)$ values (Sturm and Zibordi 2002);
4) determination of the near-surface extrapolation interval, and thus the water-leaving radiance, for those processing schemes taking advantage of the $E_d(0^+,$

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λ) and $E_d(0^-, \lambda)$ convergence to select the interval limits; and
5) derived quantities formed from ratios of $R(\lambda)$, $L_{WN}(\lambda)$, and $R_{rs}(\lambda)$ at different wavelengths.

The uncertainties in the quantities discussed in points 1–5 may be higher in radiometers with multiple diffusers (one per spectral channel) as opposed to radiometers with a single diffuser. This is due to manufacturing and production differences among diffusers, which are an inevitable source of interchannel uncertainties.

There is a requirement to produce high quality data to support ongoing validation activities for ocean color sensors (Hooker and Esaias 1993). This requirement prompted the recommendation from Mueller (1995) on the need for individual experimental characterizations of each irradiance collector. Because of the unavailability of an extensive study on the immersion factors of the OCI-200 radiometers, which are widely used in the scientific community and extensively applied in the Coastal Atmosphere and Sea Time-Series (CoASTS) program (Zibordi et al. 2002a) as well as Atlantic Meridional Transect (AMT) oceanographic cruises (Hooker and Maritorena 2000), a round-robin experiment involving different laboratories was planned and initiated.

The following were the primary objectives of the experiment:

1) the evaluation of intralaboratory repeatabilities in determining immersion factors through multiple characterizations of a so-called reference (OCI-200) radiometer;
2) the evaluation of interlaboratory uncertainties through comparisons of immersion factors for a group of OCI-200 radiometers independently characterized; and
3) the estimate of typical immersion factors and maximum uncertainty values for the irradiance collectors of the OCI-200 series of radiometers.

The secondary objectives of the experiment were

1) the intercomparison of procedures, based on the same basic protocol, used in different laboratories for determining immersion factors, and
2) the formulation of recommendations for improving immersion factor measurements.

The institutes involved in the experiment were the Center for Hydro-Optics and Remote Sensing (CHORS) of San Diego State University, the Inland and Marine Waters Unit of the Joint Research Centre (JRC), Satlantic Inc., and the National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (GSFC). The laboratory measurements took place at CHORS, JRC, and Satlantic between October and December 2001.

The experiment was called the Eighth SeaWiFS (Seaviewing Wide Field-of-view Sensor) Intercalibration Round-Robin Experiment (SIRREX-8) and complemented the incremental investigations into instrument and calibration uncertainties carried out during former SIRREX activities (Hooker et al. 2002). These activities were designed to investigate the uncertainties in the calibration equation

$$E(t, \lambda) = C_c(\lambda) I_f(\lambda)[D_N(t, \lambda) - D_0(\lambda)], \qquad (1)$$

where $E(t, \lambda)$ is the calibrated irradiance related to the digital number $D_N(t, \lambda)$ acquired at time t and wavelength λ, $C_c(\lambda)$ is the calibration coefficient determined in the laboratory during the radiometric calibration of the sensor, $I_f(\lambda)$ is the immersion factor, and $D_0(\lambda)$ is the dark value taken with the cap on the instrument, or alternatively, the background value taken with an on-axis baffle placed between the radiometer and the source to block the direct light [where $D_N(t, \lambda)$ and $D_0(\lambda)$ are both in relative units]. Although SIRREX-1 to SIRREX-7 were carried out to quantify uncertainties in $C_c(\lambda)$, SIRREX-8 focused on $I_f(\lambda)$ and was an additional step toward a complete investigation of all elements defining the calibration equation for in-water absolute radiometric data.

2. $I_f(\lambda)$ characterization

The refractive index of the diffuser material, $n_d(\lambda)$, used for manufacturing irradiance collectors, is always larger than the refractive index of water $n_w(\lambda)$ and of air $n_a(\lambda)$. Because $n_w(\lambda) > n_a(\lambda)$, the Fresnel reflectance of the external water–diffuser interface is smaller than that of the air–diffuser interface. Because of this, the transmission of light through the external interface of the diffuser is larger in water than in air. Similarly, the internal diffuser–water interface also reflects less of the internal diffuse light, when compared to the corresponding diffuser–air interface. Because of the much larger amount of light transmitted back into the water with respect to the increased amount of light transmitted into the diffuser, there is a decrease in the net irradiance measured by the detector when the instrument is in water with respect to in air.

The methodology proposed in the literature (Petzold and Austin 1988; Mueller and Austin 1995) for determining $I_f(\lambda)$ for in-water irradiance collectors is based on using the irradiances measured by the sensor in the air and just below the water surface when vertically illuminated by the same point source. In agreement with this, the immersion factor $I_f(\lambda)$ is given by

$$I_f(\lambda) = \frac{E(0^+, \lambda)}{E(0^-, \lambda)} T_s(\lambda), \qquad (2)$$

where $E(0^+, \lambda)$ is the in-air irradiance, $E(0^-, \lambda)$ is the in-water subsurface irradiance, and $T_s(\lambda)$ is the transmittance of the water surface to downward irradiance.

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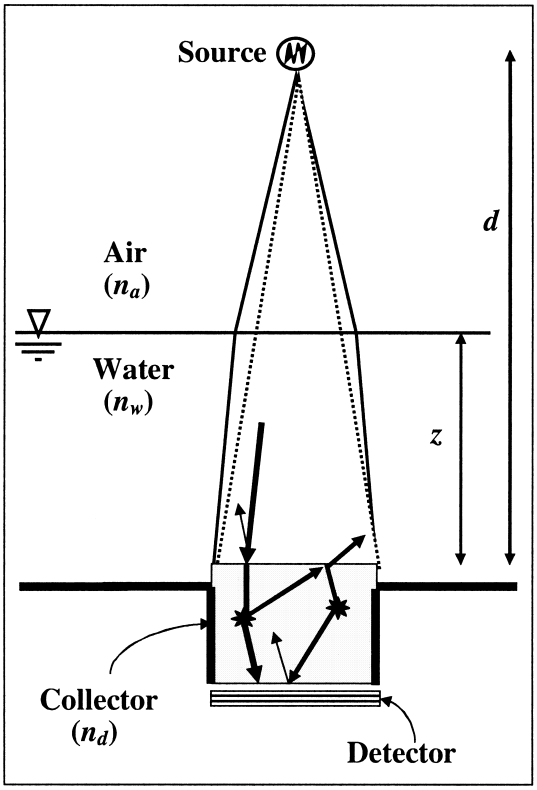

FIG. 1. A schematic of the reflection–refraction processes occurring at the medium–diffuser interface and inside an irradiance collector (assumed to be made of a single piece of diffusing material), together with the illumination geometry as induced by a finite distance, d, between the collector and the source during the measurement sequence (the dashed line shows the in-air illumination geometry, while the continuous line highlights the illumination geometry with water depth, z, above the collector).

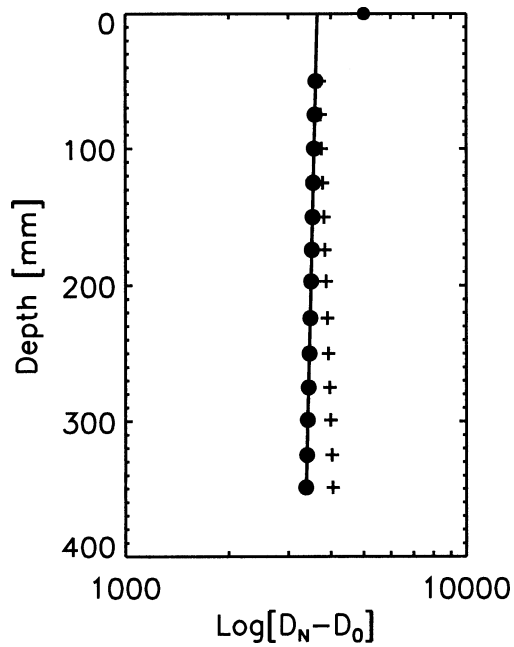

FIG. 2. Sample data for E_u130, used for $I_f(\lambda)$ characterization at the 555-nm center wavelength. Crosses indicate actual measurements at different water depths. Dots at depth $z > 0$ indicate data corrected for geometric effects [through the $G(z, \lambda)$ factor]. The dot at zero depth indicates the in-air measurement to be divided by the subsurface extrapolated value (highlighted by the intercept of the regression line).

The latter is computed from Fresnel reflectance for a vertically incident light beam and given by

$$T_s(\lambda) = \frac{4n_w(\lambda)}{[1 + n_w(\lambda)]^2}. \tag{3}$$

The characterization of $I_f(\lambda)$ requires the computation of $E(0^-, \lambda)$ from the least squares fit of the logarithm of in-water irradiance data [determined from (1) with $I_f(\lambda) = 1$] corrected by the measurement perturbations induced by the finite distance between an ideal point source and the collector, that is, $\ln[E(z, \lambda)/G(z, \lambda)]$, where $G(z, \lambda)$ is the so-called geometric correction factor. The $G(z, \lambda)$ values for the irradiance $E(z, \lambda)$ with a point source at distance d from the collector are computed according to Aas (1969) and Petzold and Austin (1988):

$$G(z, \lambda) = \left\{1 - \frac{z}{d}\left[1 - \frac{1}{n_w(\lambda)}\right]\right\}^{-2}, \tag{4}$$

where values of $n_w(\lambda)$ in the spectral range 400–700 nm for pure water and pure seawater (i.e., with a salinity of 0 PSU and 35 PSU, respectively, both at a temperature of 20°C) can be obtained by

$$n_w(\lambda) = 1.318\,91 + \frac{6.314\,46}{\lambda - 139.596} \quad \text{and} \tag{5}$$

$$n_w(\lambda) = 1.324\,83 + \frac{6.533\,18}{\lambda - 139.589}. \tag{6}$$

The coefficients given in (5) and (6) were computed by fitting tabulated data from Austin and Halikas (1976) with λ in nanometers (Zibordi et al. 2002b). Equivalent equations were already proposed by Petzold and Austin (1988) for pure water at 22°C and by Mueller and Austin (1995) for pure seawater at 16°C.

A schematic of the major reflection–refraction processes occurring within the diffuser and at the diffuser interfaces, together with a schematic of the measurement geometry applied for $I_f(\lambda)$ characterization, are given in Fig. 1. Sample data produced in agreement with the described methodology are displayed in Fig. 2. They highlight the relevance of the geometric correction of the irradiance data taken with different water depths.

3. Intra- and interlaboratory comparison experiments

The in-air $E(0^+, \lambda)$ and in-water $E(z, \lambda)$ measurements were obtained by averaging data collected during a sampling interval (typically ranging from 1.5 to 3 min,

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TABLE 1. The principal features of the systems used for characterizing $I_f(\lambda)$ at the different laboratories (dimensions are indicated by ⌀ for the diameter, H for the height, W for the width, and L for the length).

Feature	CHORS	JRC	Satlantic
Lamp type	64663 (Osram)	64575 (Osram)	FEL (Sylvania)
Applied lamp power (W)	330	770/870[a]	1000
Lamp filament (cm)	0.4 × 0.8	0.3 × 1.0 (× 3)[b]	0.6 × 2.0
Lamp–sensor distance (cm)	86	105	100
Primary baffle aperture (cm)	5.0 (⌀)	8.5 (⌀)	25 (W) × 25 (L)
Primary baffle–sensor distance (cm)	53.5	39.0	59.0
Lamp screen aperture (cm)	3.5 (⌀)	4.6 (⌀)	—
Lamp screen–lamp distance (cm)	3.5	6.3	—
Tank size (cm)	152 (H) × 81 (⌀)	70 (H) × 80 (⌀)	123 (H) × 120 (W) × 243 (L)
Volume of water (L)	780[c]	350[d]	3500[e]
Max depth of water (cm)	40	35	50
Min depth of water (cm)	5	5	5
Depth increment (cm)[f]	2.5	2.5	5
Measurement depths	15	13	10
In-air measurements	2	1	1
Dark and background measurements	2	2	1
Recording time per depth (min)	3	1.5	2
Execution time per sequence (min)	120	100	330[g]

[a] Used for the characterization of E_u and E_d radiometers, respectively.
[b] Composed of three interconnected filaments (⊓ shaped).
[c] Tap water.
[d] Filtered tap water recycled for successive characterizations.
[e] Aged seawater recycled, chlorinated, and filtered for successive instrument characterizations.
[f] Increment in water depth at successive measurements with 0.5-mm accuracy.
[g] Including at least 3 h of water resting to reduce the number of bubbles.

according to procedures implemented at the different laboratories) and normalized with respect to data from a radiometer monitoring the source. The normalization is used to reduce uncertainties caused by changes in the flux of the light source during the measurement sequence. The in-air and the in-water data, after removing dark (or background) data, are divided by the corresponding (temporally matched) data recorded from the monitoring radiometer, and are then multiplied by the monitoring radiometer data taken at initial time t_0, corresponding to the start of the measurement sequence.

The elements used to evaluate data quality are (i) the standard deviation σ for the data collected in air or at each specific water depth during the defined sampling interval; (ii) the negative values of slopes from the least squares fit used to compute $E(0^-, \lambda)$, which are denoted here as $K(\lambda)$; and (iii) the percent difference between the actual values $E(z, \lambda)/G(z, \lambda)$ and the fitted $E(0^-, \lambda)\exp[-K(\lambda)z]$ values at each depth z. A high σ value suggests changes in the measurement conditions during one step of the data collection (i.e., due to an instability of the water surface or the presence of large particles floating above the collectors). As a consequence of using a light source that approximates a direct beam illuminating the diffusers, the $K(\lambda)$ value is a function of the water absorption and scattering, and of the optical–mechanical setup. Significant differences in $K(\lambda)$ among successive measurement sequences, therefore, imply changes in water quality or in the geometry of the system. Significant changes in the percent difference of $E(z, \lambda)/G(z, \lambda)$ and the fitted $E(0^-, \lambda)\exp[-K(\lambda)z]$, at a specific depth z, may indicate an incorrect determination of the depth z or changes in the optical setup during the specific measurement sequence.

a. *Measurement implementation at the different laboratories*

The protocol for $I_f(\lambda)$ characterization, as described in Mueller and Austin (1995), was independently implemented (with different lamp types and related geometries, mechanical setups, tank sizes, and water types) at the three different laboratories. A detailed description of each implementation is given in Zibordi et al. (2002b). For completeness, the light source characteristics and geometry, together with other relevant features defining the measurement procedures, are summarized in Table 1 for each laboratory. As an example, the schematic of the JRC $I_f(\lambda)$ measurement setup is shown in Fig. 3.

The major difference between laboratories, besides the different implementations of a point source, was the type of water used and the techniques adopted to ensure its purity (including how the surface was kept free of particles). CHORS used tap water partly renewed for each measurement sequence. JRC used filtered tap water (with a resistivity greater than 5–8 MΩ) recycled over several measurement sequences through a temporary storage tank. Satlantic used seawater filtered through a series of sand filters with chlorine added and aged more than 1 yr in the laboratory, recycled through a temporary storage tank, and subject to coarse (about 1 μm) filtra-

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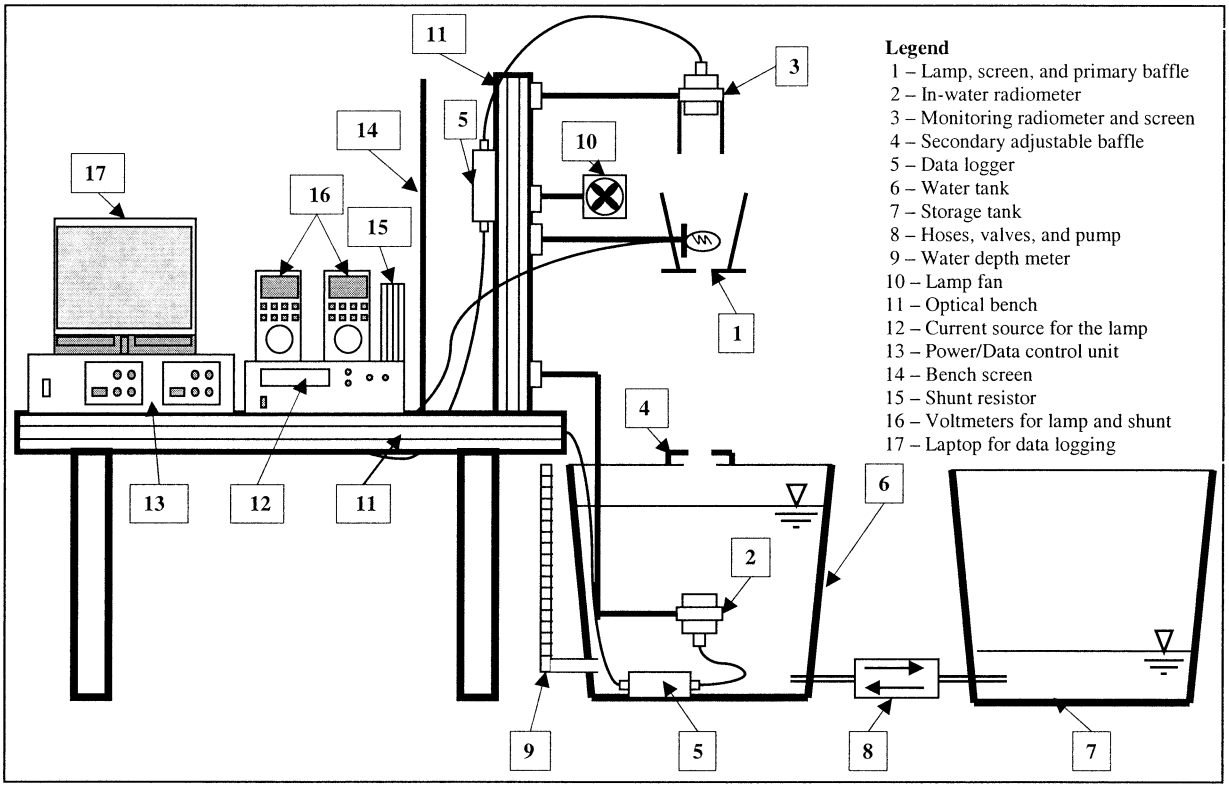

FIG. 3. A schematic of the JRC $I_f(\lambda)$ measurement setup.

tion for each measurement sequence. CHORS and JRC skimmed the surface to remove floating particles, whereas Satlantic used small quantities of soap to reduce surface tension, which removed particles in the area illuminated by the source. The use of soap is based on the assumption that a small quantity of soap spreads over the surface in a monomolecular layer, and that in the limiting case of a layer thickness much smaller than the wavelength, it does not modify the transmittance of the water surface (and consequently does not perturb the measurement).

An additional element differentiating the water used at the three laboratories was the temperature. At JRC and Satlantic, the water temperature was always close to ambient (i.e., 20°–22°C). At CHORS, because of the continuous renewal of a portion of the water in the tank, the water temperature was generally lower than ambient (with an estimated value of approximately 18°C), but fairly constant over time.

The optical alignment between the lamp and the leveled radiometer was made using a laser at JRC and at Satlantic. At CHORS, it was made by visually centering the radiometer within the light cone projected by the source to the inside of the tank. An additional effort was made at JRC to preserve the rotational orientation of the radiometers and ensure accurate repositioning during successive characterizations, by using the flat edge of D-shaped collars attached to the radiometers.

The common feature to all three laboratories was the use of a radiometer [i.e., an OCI-200 designed for $E_d(0^+, \lambda)$ measurements] to monitor the stability of the light source, added to the use of a precision shunt resistor in series with the lamp, to independently detect changes in lamp flux during each step of the measurement sequences.

b. *The series of radiometers considered in the analysis*

The intercomparison experiment was restricted to the widely used seven-channel OCI-200 radiometers. These radiometers were designed to detect light over a four-decade range using 16-bit analog-to-digital conversion. The seven channels are associated with independent diffusers manufactured by layering different acrylic materials. Each diffuser has scattering and absorption properties optimized for the specific center wavelength. The diffusers are symmetrically arranged on the radiometer faceplate, with the seventh diffuser located in the center. The optics and electronics are housed in a cylinder with a diameter of 8.9 cm and a length of 11.2 cm.

A total of nine OCI-200 radiometers (four E_u and five E_d sensors), all having the same nominal center wavelengths (i.e., 412, 443, 490, 510, 555, 665, and 683 nm), were included in the experiment. The selected radiometers were manufactured between September 1994

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TABLE 2. The number of quality-assured vs total (in parentheses) measurement sequences carried out at each laboratory for each OCI-200 radiometer (identified by type, serial number, and manufacture date).

Radiometer, SN (Manufacture date)	CHORS	JRC	Satlantic
E_d 015 (Sep 1994)	1 (2)	2 (2)	2 (2)
E_d 040 (Mar 1996)	1 (2)	1 (2)	2 (2)
E_u 048 (Jun 1996)	3 (4)	2 (3)	2 (2)
E_d 050 (Jun 1996)	1 (2)	1 (2)	2 (2)
E_d 071 (Apr 1997)	1 (2)	3 (4)	2 (2)
E_d 097 (Jun 1998)	1 (2)	2 (3)	1 (2)
E_u 098 (May 1998)	1 (2)	1 (2)	2 (2)
E_u 109 (Jul 1998)	1 (2)	3 (4)	2 (2)
E_u 130* (Jul 1999)	5 (8)	7 (10)	6 (6)
Total	15 (26)	22 (32)	21 (22)

* Reference radiometer.

and July 1999 and represent about 10% of the production of the in-water OCI-200 series of radiometers for the considered manufacturing time period. One E_u sensor (E_u130) was used as a reference unit, and was frequently characterized to (a) monitor the stability of the entire measurement system over the course of the experiment and (b) evaluate the method precision to establish whether a single radiometer characterized many times provides the same statistical results as several radiometers characterized a small number of times. The number of radiometers included in the experiment was chosen to (a) ensure both the significance of the statistical analysis and (b) permit the execution of this study without negatively impacting previously scheduled field and laboratory activities. A larger number of radiometers and characterizations would have made the experiment more statistically relevant, but its execution would have not been possible in the period imposed by the aforementioned work schedules.

4. The dataset

The number of measurement sequences included in the present data analysis are listed in Table 2 for each laboratory. In several cases, only one measurement sequence per laboratory was retained for the intercomparison experiment after the data were quality assured.

Some of the CHORS data were affected by the presence of large particles floating on the surface. This perturbation, when not removed, produces a decrease in the transmittance of the air–water surface that induces an increase in the computed $I_f(\lambda)$. Measurement sequences from CHORS were quality assured after introducing surface skimming (made before the start of the data acquisition sequences) in the measurement protocol. Some of the JRC data were affected by a decrease in water purity, probably caused by particles falling into the water or released by the pump, hoses, and tanks. This perturbation produces a decrease in computed $I_f(\lambda)$ values because of an increase in light scattering. Quality assurance of JRC data was carried out on the basis of a threshold (i.e., 0.2 m^{-1}) applied to computed $K(412)$ values. One measurement sequence from Satlantic was affected by perturbations in the optical–mechanical setup during its execution and was not retained.

The applied quality assurance for the removal of data affected by well-identified perturbing factors aims at reducing sources of bias.

5. Data analysis

All quality-assured data from the three laboratories were processed using the same software to remove uncertainties caused by different parameter settings and approximations inherent to any processing scheme. CHORS and JRC data were processed using the refractive index of pure water (5), whereas Satlantic data were processed using the refractive index of pure seawater (6).

The intercomparison among $I_f(\lambda)$ values produced with measurement sequences from the three laboratories was made through the analysis of $I_f(\lambda)$ data from the E_u130 reference radiometer and those from all radiometers. The $I_f(\lambda)$ data from the E_u130 reference radiometer provide an intercomparison among the different laboratories and supports an intralaboratory analysis of the measurement repeatability (the number of measurement sequences for E_u130 was the largest). The $I_f(\lambda)$ data from all the radiometers, in addition to providing intercomparability among the different laboratories, permit an analysis of the dispersion of $I_f(\lambda)$ data across the OCI-200 sample radiometers covering the 1994–99 production time period.

Results from the data analysis are presented in the following sections. The data are displayed as a function of the nominal center wavelengths (which were the same for all the radiometers).

a. $I_f(\lambda)$ data analysis for the reference radiometer

The results of the intra- and interlaboratory analyses for the $I_f(\lambda)$ values of the reference E_u130 radiometer computed with quality-assured measurements are presented in Table 3. Measurement repeatability is quantified through the parameter $\xi(\lambda)$ given by two times the standard deviation $\sigma(\lambda)$ divided by the average $\bar{I}_f(\lambda)$ of independent $I_f(\lambda)$ measurements, and expressed as a percentage; that is, $\xi(\lambda) = 200\sigma(\lambda)/\bar{I}_f(\lambda)$. Superscripts RC, RJ, and RS are used to indicate values for the reference radiometer produced with CHORS, JRC, and Satlantic measurements, respectively.

The relative interlaboratory uncertainties are presented through the unbiased percent difference $\psi(\lambda)$ (i.e., the percent difference between two quantities with respect to their average) between the spectral average $\bar{I}_f(\lambda)$ data from each laboratory [i.e., $\bar{I}_f^{RC}(\lambda)$, $\bar{I}_f^{RJ}(\lambda)$, and $\bar{I}_f^{RS}(\lambda)$] and the average values $\bar{I}_f^{RA}(\lambda)$ computed with the averages from the three laboratories

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TABLE 3. The intra- and interlaboratory comparison of $I_f(\lambda)$ data for the E_u130 reference radiometer. The analysis, supported by N-independent measurements (N = 5, 7, and 6 for CHORS, JRC, and Satlantic, respectively), is presented for each laboratory as a function of λ. Superscripts RC, RJ, and RS indicate values for the reference radiometer produced with CHORS, JRC, and Satlantic measurements, respectively.

λ (nm)	ξ^{RC} (%)	ξ^{RJ} (%)	ξ^{RS} (%)	ξ^{RA} (%)	ψ^{RC} (%)	ψ^{RJ} (%)	ψ^{RS} (%)
412	0.71	0.75	0.42	1.47	−0.75	0.03	0.72
443	0.72	0.45	0.20	1.27	−0.62	−0.04	0.65
490	0.61	0.32	0.27	0.73	−0.36	−0.02	0.38
510	0.59	0.34	0.26	1.06	−0.53	0.00	0.53
555	0.57	0.24	0.44	1.21	−0.55	−0.09	0.64
665	0.50	0.73	0.17	1.16	−0.39	−0.28	0.67
683	0.48	0.59	0.18	1.33	−0.51	−0.25	0.75
Avg	0.60	0.49	0.28	1.18	−0.53	−0.09	0.62

[i.e., $\bar{I}_f^{RA}(\lambda) = [\bar{I}_f^{RC}(\lambda) + \bar{I}_f^{RJ}(\lambda) + \bar{I}_f^{RS}(\lambda)]/3$]. The method precision is then determined by $\xi^{RA}(\lambda)$ [computed in agreement with the defined $\xi(\lambda)$] with the $\bar{I}_f^{RA}(\lambda)$ values.

The $I_f(\lambda)$ intralaboratory analysis for the reference E_u130 radiometer shows the highest measurement repeatability for Satlantic with an average $\xi^{RS}(\lambda)$ of 0.28%. JRC and CHORS display average values of $\xi^{RJ}(\lambda)$ and $\xi^{RC}(\lambda)$ of 0.49% and 0.60%, respectively. Typically, the lowest repeatability is observed at 412 nm for all three laboratories.

The interlaboratory comparison, based on the E_u130 $I_f(\lambda)$ data, shows a method precision $\xi^{RA}(\lambda)$ ranging from 0.73% to 1.47% with an average value of 1.18%. The highest value of $\xi^{RA}(\lambda)$ is observed at 412 nm (1.47%), while the lowest value (0.73%) is observed at 490 nm. The relative interlaboratory uncertainties determined with the average E_u130 $I_f(\lambda)$ values from each laboratory and the average of data from all three laboratories, $\bar{I}_f^{RA}(\lambda)$, show the lowest values for JRC with $\psi^{RJ}(\lambda)$ ranging from −0.25% to 0.03% with an average of −0.09%. CHORS and Satlantic show relative uncertainties similar in magnitude but with opposite sign. Specifically CHORS displays $\psi^{RC}(\lambda)$ values ranging from −0.75% up to −0.36% with an average of −0.53%. Satlantic displays $\psi^{RS}(\lambda)$ values ranging from 0.38% up to 0.75% with an average of 0.62%.

b. $I_f(\lambda)$ data analysis for the OCI-200 series of radiometers

The intra- and interlaboratory analysis data resulting from the characterization of the nine OCI-200 radiometers included in the experiment are shown Table 4. When multiple $I_f(\lambda)$ measurements were available for a specific radiometer, the average was used in the analysis. The superscripts AC, AJ, and AS are used to indicate average values for all nine radiometers from CHORS, JRC, and Satlantic, respectively.

The dispersion of the $I_f(\lambda)$ measurements across the nine radiometers is quantified through the parameter $\xi(\lambda)$ determined from the average $\bar{I}_f(\lambda)$ of the $I_f(\lambda)$ measurements of each radiometer and the related standard deviation $\sigma(\lambda)$.

Similar to the comparison for $I_f(\lambda)$ data from the reference E_u130 radiometer, the relative interlaboratory uncertainties for all nine radiometers are presented through unbiased percent differences, $\psi(\lambda)$, between the spectral average data from each laboratory (i.e., $\bar{I}_f^{AC}(\lambda)$, $\bar{I}_f^{AJ}(\lambda)$, and $\bar{I}_f^{AS}(\lambda)$) and average values $\bar{I}_f^{AA}(\lambda)$ computed from the averages from the three laboratories {i.e., $\bar{I}_f^{AA}(\lambda) = [\bar{I}_f^{AC}(\lambda) + \bar{I}_f^{AJ}(\lambda) + \bar{I}_f^{AS}(\lambda)]/3$}. The precision of the method $\xi^{AA}(\lambda)$ is quantified from the $\bar{I}_f^{AA}(\lambda)$ values.

The intralaboratory analysis exhibits average values of the $I_f(\lambda)$ dispersion over the nine OCI-200 radiometers for the three laboratories, $\xi^{AC}(\lambda)$, $\xi^{AJ}(\lambda)$, and $\xi^{AS}(\lambda)$, ranging from 2.2% to 2.3%. Instrument-to-instrument variability, however, may be quite spectrally pronounced. Data from all three laboratories display high variability in $I_f(\lambda)$ at 665 and 490 nm with dispersion values of 5% and 3% (defined by 2σ), respectively (mostly associated with outliers from E_d015). The lowest dispersion is observed at 510 nm, with values generally lower than 1%.

TABLE 4. The intra- and interlaboratory comparison of $I_f(\lambda)$ data for the nine OCI-200 radiometers included in the intercomparison experiment. The analysis is presented for each laboratory as a function of λ. Superscripts AC, AJ, and AS indicate values for the reference radiometer produced with CHORS, JRC, and Satlantic measurements, respectively.

λ (nm)	ξ^{AC} (%)	ξ^{AJ} (%)	ξ^{AS} (%)	ξ^{AA} (%)	ψ^{AC} (%)	ψ^{AJ} (%)	ψ^{AS} (%)
412	1.78	1.69	1.48	1.23	−0.52	−0.16	0.68
443	1.20	1.30	1.23	1.22	−0.54	−0.13	0.66
490	2.97	3.02	2.57	0.94	−0.45	−0.05	0.49
510	1.04	0.74	0.92	1.24	−0.55	−0.12	0.66
555	1.68	0.89	1.48	1.37	−0.54	−0.23	0.77
655	4.95	5.50	5.12	1.08	−0.36	−0.26	0.62
683	2.47	2.45	2.57	1.07	−0.30	−0.31	0.61
Avg	2.30	2.23	2.20	1.16	−0.47	−0.18	0.64

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The interlaboratory comparison, based on $I_f(\lambda)$ data from all nine OCI-200 radiometers, shows an average method accuracy, $\xi^{AA}(\lambda)$, of 1.16% displaying the highest values at 555 nm (1.37%) and the lowest (0.94%) at 490 nm. The relative interlaboratory uncertainties show the lowest values for JRC with $\psi^{AJ}(\lambda)$ ranging from −0.31% to 0.05% and an average of −0.18%. CHORS displays values of $\psi^{AC}(\lambda)$ ranging from −0.55% to −0.30% and an average of −0.47%. Satlantic displays values of $\psi^{RS}(\lambda)$ ranging from 0.49% to 0.77% with an average of 0.64%. All of the former data confirm the interlaboratory comparison results produced with $I_f(\lambda)$ data from the E_u130 reference radiometer (Table 3).

6. Discussion

The comparisons show different interlaboratory repeatabilities and interlaboratory relative uncertainties in $I_f(\lambda)$ characterization. In the following subsections, the sources of the observed differences are discussed together with the major sources of uncertainty associated with the characterization of $I_f(\lambda)$ with the applied methodology. In addition, typical OCI-200 $I_f(\lambda)$ values are proposed with their maximum estimated uncertainties to support the processing of data collected with OCI-200 radiometers not individually characterized. Finally, an analysis of the $I_f(\lambda)$ values is presented together with a description of the processes determining the spectral dependence.

a. Intra- and interlaboratory uncertainties

Immersion factors $I_f(\lambda)$ computed from measurements carried out at Satlantic showed the highest repeatability (i.e., with average $\xi^{RS} = 0.28\%$). This result can probably be explained by the good alignment procedure and the high stability of the water used in the experiment (i.e., filtered seawater with chlorine added, aged for at least 1 yr in a laboratory tank, and subjected to coarse and repeated filtration). The relatively higher variability in $I_f(\lambda)$ data from measurements performed at CHORS (with average $\xi^{RC} = 0.60\%$) and JRC (with average $\xi^{RJ} = 0.49\%$) can be attributed to different sources. For CHORS, it may be justified by possible slight differences in the alignment of radiometers with respect to the source (made visually) and by changes over time in the quality of the tap water. For the JRC, because of the good optical alignment, it can only be attributed to changes in water quality over time (due to dust particles falling in the water and the release of particles from the pump, hoses, and tanks).

The relative interlaboratory accuracies shown by the analysis of E_u130 data (Table 3) and confirmed by the analysis of all nine OCI-200 radiometers (Table 4) has to be investigated in terms of differences in the optical–mechanical setups, and the quality and type of water available at the three different laboratories. Assuming negligible perturbations in $I_f(\lambda)$ characterization by the reflection of the walls of the tanks and by the mechanical supports used for mounting the in-water radiometer, the explanation for systematic differences should be investigated by considering (i) the presence of scattering material; (ii) the use of tap water versus seawater (the latter required for the marine use of the radiometer), both of which produce an underestimate of $I_f(\lambda)$; and (iii) the existence of slicks at the surface, which produce an overestimate of $I_f(\lambda)$. To support the former hypothesis, analysis of absorption coefficients for particulate $a_p(\lambda)$ and dissolved $a_y(\lambda)$ matter were made on water samples from the three laboratories. Water samples from JRC and Satlantic were taken from the laboratory tanks during the experiments. The water sample from CHORS, was taken from the tap about 3 months after the experiment (the latter absorption data are used here on the assumption that the tap water available at CHORS had stable optical properties over time).

TABLE 5. The particulate $a_p(\lambda)$ and dissolved $a_y(\lambda)$ matter absorption coefficients at λ = 412 and 650 nm, measured for water samples taken at the three laboratories.

Absorption coef (m^{-1})	CHORS	JRC	Satlantic
$a_p(412)$	0.142	0.019	0.059
$a_p(650)$	0.075	0.002	0.008
$a_y(412)$	0.124	0.149	0.253
$a_y(650)$	0.000	0.062	0.000

The absorption coefficients for particulate matter given in Table 5 show the highest value (0.142 m^{-1} at 412 nm) for CHORS and the lowest value (0.019 m^{-1} at 412 nm) for JRC. The particulate absorption coefficient for the Satlantic water sample shows intermediate values (0.059 m^{-1} at 412 nm). Even though the absorption coefficient per se should not significantly affect $I_f(\lambda)$ estimates, the former $a_p(\lambda)$ values can be an indicator, to a first approximation, of the presence of scattering particles inducing an underestimate of $I_f(\lambda)$. Using this hypothesis, the high particulate absorption coefficient (assumed as an indicator of a high concentration of scattering particles) may explain the lowest $I_f(\lambda)$ values obtained for CHORS. This does not, however, directly explain the Satlantic $I_f(\lambda)$ values being higher than the JRC values. One reason for this apparent inconsistency comes from the analysis of the absorption coefficients from dissolved matter $a_y(\lambda)$ and from differences in the measurement protocol.

Under the assumption that the material does not scatter, absorption $a_y(\lambda)$, defined here by material with a diameter of less than 0.22 μm, should not have any appreciable effect on $I_f(\lambda)$. This hypothesis is generally supported by negligible absorption at 650 nm, fully confirmed for the CHORS and Satlantic samples, but not for JRC samples. This finding, combined with the analysis of the whole absorption spectrum of $a_y(\lambda)$, suggests

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the presence of colloidal matter that might act as a scatterer, which would produce an underestimate of the JRC $I_f(\lambda)$ values. A further element explaining the systematic differences observed across laboratories comes from the use of different water types and from the methods applied to remove surface particles. Assuming the same level of purity, the use of tap water versus seawater (characterized by slightly different refractive indices) introduces a negative bias in the CHORS and JRC immersion factors when compared to the Satlantic values. In addition, while CHORS and JRC skimmed the water surface, Satlantic made use of small quantities of soap to reduce surface tension in the area where the measurements were made. The latter method is quite efficient with seawater and provides quick and stable removal of surface particles. The slick forming at the surface, however, could change the water surface reflectance, thereby reducing the accuracy of (3), and become a source of uncertainty leading to an overestimate of $I_f(\lambda)$.

The former elements provide a tentative explanation for the systematic differences across the three laboratories, but they do not exclude the presence of further unexplored sources of uncertainty (e.g., microbubbles, films on the diffusers, the use of chlorine) that may affect the absolute characterization of $I_f(\lambda)$.

b. Uncertainties in $I_f(\lambda)$ characterization

The intercomparison results from this study show that the uncertainties in the characterization of $I_f(\lambda)$ for irradiance collectors, excluding any contribution from the optical–mechanical setup, are mostly sensitive to (a) the level of water purity, (b) the presence of dust particles or slicks by water contaminants (e.g., silicon grease released by connectors, soap used to reduce the tension of water surface) at the water surface, and (c) the use of tap water versus seawater.

Additional elements, which could become a source of uncertainty, if improperly handled, are (d) the assumption of a point source for the lamp, (e) the optical load of the radiometer (i.e., the perturbation induced in measurements by the high reflectance of diffusers), (f) the stability of the light source, and (g) the water temperature.

All these sources of uncertainty may produce additive effects increasing or reducing, in an almost unpredictable way, the total uncertainty in the $I_f(\lambda)$ characterization.

Specifically, purity of water is the most difficult accomplishment in $I_f(\lambda)$ characterization especially when large volumes of water are required. As already pointed out, absorption processes do not significantly affect the determination of the subsurface irradiance. Contrarily, the presence of scattering elements may produce an underestimate of $I_f(\lambda)$, because of an increase in the subsurface irradiance induced by an increase in the irradiance contribution reflected back into the water by the water–air interface. The analysis of $I_f(\lambda)$ data produced in the present study with the CHORS and JRC measurements taken with tap water (assuming the JRC data are not significantly affected by any particulate or dissolved matter scattering in water and the CHORS data are only significantly affected by particulate matter scattering) shows that perturbations due to scattering are more pronounced in the blue part of the spectrum and decrease toward the red (in agreement with theoretical expectations). For the specific case considered here (Table 5), the underestimate attributed to particle scattering varies from approximately 0.8% at 412 nm to 0.1% at 665 nm.

The presence of particles or slicks at the water surface can induce an underestimate of subsurface irradiance that leads to an underestimate of $I_f(\lambda)$ that is almost constant across the visible spectrum. Surface skimming and care in the use of parts of the setup that may cause contamination can easily minimize the effects of this potential source of uncertainty. An extreme case of surface contamination by floating particles observed at CHORS showed an average overestimate of 1.6% in $I_f(\lambda)$. The use of soap was seen to be quite efficient in reducing surface tension in seawater and, thus, in moving particles away from the surface area directly associated with the optical measurements. No specific experiment, however, was executed to investigate possible perturbations in the surface transmittance due to the presence of a soap slick.

The use of tap water versus seawater introduces a negative bias in the absolute characterization of immersion factors. This bias is expected to be of the same order of the difference in refractive indexes of pure water and pure seawater (i.e., about 0.5% in the 400–700-nm spectral range) and is smaller than the differences observed between the JRC and Satlantic immersion coefficients determined with tap water and seawater, respectively. In fact these differences, ranging from 0.6% in the blue to 1.0% in the red, also account for the presence of water surface contaminants and for different water purities. The use of a point source, (i.e., a lamp), at a finite distance from the collector, induces different illumination conditions on the collector as a function of the changing water depth, which leads to an overestimate of $I_f(\lambda)$. Combining the use of small filament lamps with increased distances between the lamp and collector, in addition to the use of the geometric correction model for in-water radiometric data as proposed by Aas (1969) and later by Petzold and Austin (1988), minimizes the uncertainties in determining $I_f(\lambda)$. A specific experiment showed that, with a lamp filament of 4 mm $\times$ 8 mm at two distances from the collector (at 86 cm and at 100 cm), the difference in $I_f(\lambda)$ values computed by applying the geometric factor $G(z, \lambda)$ (4) was within the intralaboratory measurement repeteability. For this experiment, percent corrections produced in $I_f(\lambda)$ by applying the factor G were -0.6% and -0.5% from 412 to 555 nm for the 86- and 100-

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cm distances, respectively. Corrections were negligible in the red channels because of the relatively high attenuation of water (which makes the extrapolation of the subsurface irradiance values less dependent on slight changes of the illumination geometry).

The number, size, and high reflectance of the irradiance collectors are additional perturbation elements in the $I_f(\lambda)$ characterization. In fact, the collectors, especially at distances close to the water surface (i.e., within a few centimeters), induce overestimates in the in-water irradiance, which are caused by multiple reflections between the immersed collectors and the air–water surface, which results in an underestimate of $I_f(\lambda)$. Berger (1958) showed that the perturbation becomes negligible when the water depth is greater than the critical depth $z_c \approx 0.9\ r_d$, where r_d is the radius of the collector. Assuming that for the OCI-200 radiometers the perturbation is produced by the area covered by all seven diffusers (i.e., a circular area of radius $r_d \approx 2.0$ cm), a depth $z_c \approx 1.8$ cm satisfies the requirement for minimizing the perturbation. In the intercomparison experiment, a critical depth $z_c \approx 5$ cm (i.e., larger than the radius of the radiometer case) was used at all laboratories. A specific measurement sequence including in-water data collected with 1-cm-depth increments between 5- and 1-cm depth, in addition to regular depth measurements beyond a depth of 5 cm, showed an average $I_f(\lambda)$ underestimate ranging between 0.2% in the blue and up to 0.9% in the red. The latter values decreased below the intralaboratory measurement repeatability when only data at depths $z \geq 2$ cm (i.e., $z > z_c$) were included in the $I_f(\lambda)$ computations.

Monitoring the stability of the source, either by using a shunt resistor in series with the lamp or a radiometer directly looking at the source, is a good practice to ensure detection (with the shunt only) and minimization (with the monitoring radiometer) of any change in the illumination conditions during each measurement sequence. During the intercomparison experiment, the high stability of the source used at all three laboratories induced differences generally lower than 0.1% in $I_f(\lambda)$ values computed by normalizing, and by not normalizing, the in-air and in-water irradiance data, with respect to the data from the monitoring radiometer.

Changes in the water temperature, which affect the seawater refractive index, are also a source of uncertainty. An estimate of the effects induced by a change in the water refractive index associated with a 4°C variation in water temperature suggests variations within 0.1% in the $I_f(\lambda)$ values.

All of the former sources of uncertainty can be easily minimized by creating a good point source and monitoring its stability, avoiding collection of data affected by the optical load of the collectors, keeping the surface free of particles and slicks, and finally using the purest water possible. Under the same conditions of water purity, the use of seawater is preferred over the use of freshwater because of the slight difference in the refractive index (i.e., approximately 0.5% in the spectral range 400–700 nm). The production of large quantities of pure seawater requires much more effort than the production of large quantities of pure water, so the requirement of determining highly accurate immersion coefficients for marine radiometers suggests the use of pure water with the application of a correction term for seawater to account for differences in the refractive indices.

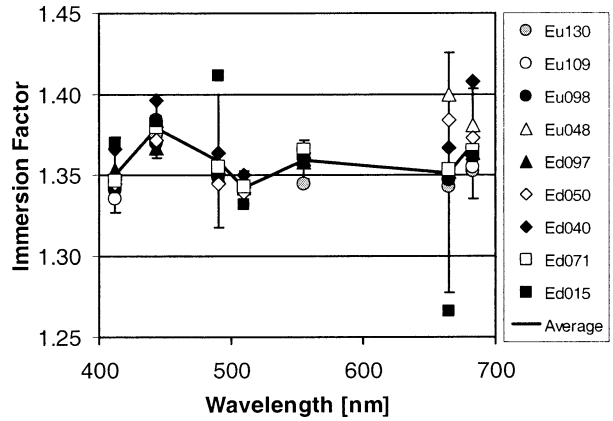

FIG. 4. The spectral JRC $I_f(\lambda)$ data for the nine OCI-200 radiometers included in the experiment. The continuous thick line highlights the average spectral $I_f(\lambda)$ values and the bars indicate $\pm 2\sigma$.

c. *Typical $I_f(\lambda)$ values for the OCI-200 series of radiometers*

Figure 4 presents the JRC spectral $I_f(\lambda)$ values for each of the nine OCI-200 radiometers included in the experiment and the resulting average $I_f(\lambda)$ values (i.e., $\bar{I}_f^{AJ}(\lambda)$ used to compute the data in Table 4). Although most of the data show good agreement among different sensors, there are notable outliers at 490 and 665 nm, which are associated with the E_d015 radiometer. Additional radiometers exhibiting relevant differences from the average are E_u048 and E_d050 at 665 nm, and E_d040 at 683 nm. These radiometers are the oldest among those included in the experiment, and a visual inspection indicated no significant damage of the collectors (as expected because of the utmost care given to the radiometers during shipping and field use). Removing the E_d015 data, the dispersion of the $I_f(\lambda)$ values defined by 2σ goes from 3.02% and 5.50% to 0.86% and 2.96% at 490 and 665 nm for the JRC data.

Excluding E_d015 and partitioning the radiometers in two subsets of new (E_d097, E_d098, E_u109, and E_u130) and old (E_d040, E_u048, E_d050, and E_d071) devices, the $I_f(\lambda)$ values show a general increase with aging. For the considered subsets, the average increase is 0.9%, with individual extreme values of up to 2% in the red (notably, the 510-nm channel does not show any significant change). Even though no final conclusion can be drawn from the former results because of the small

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TABLE 6. The typical immersion coefficients resulting from this study for the OCI-200 series of radiometers; I_f^O (λ) data are for the subset of old radiometers, I_f^N (λ) data are for the subset of new radiometers, and I_f^T (λ) data are for the whole set of trusted radiometers (i.e., E_d015 excluded). The maximum uncertainties are the sum of the average intralaboratory measurement repeatability and $I_f(\lambda)$ dispersion values across the considered OCI-200 radiometers (E_d015 excluded).

λ (nm)	I_f^O	I_f^N	I_f^T	Max uncertainties (%)
412	1.360	1.349	1.355	±2.1
443	1.388	1.381	1.385	±2.1
490	1.362	1.354	1.358	±1.5
510	1.351	1.350	1.350	±1.4
555	1.371	1.363	1.367	±2.3
665	1.384	1.355	1.370	±3.4
683	1.391	1.367	1.379	±3.1

statistical representation of the considered samples, some difference over time in the manufacturing process of collectors or changes of the optical features with aging (probably due to their use) are suggested, which is more pronounced in the red part of the spectrum (i.e., at 665 and 683 nm).

The averaging of values across laboratories is considered a viable solution for preventing any biasing in the results, because (i) all the laboratories involved in the experiment used a method satisfying the basic protocol for determining the immersion factor and (ii) the interlaboratory $I_f(\lambda)$ uncertainties are much lower than the spectral dispersion due to differences across radiometers (i.e., on average lower than 1% for the former and greater than 2% for the latter). The typical $I_f(\lambda)$ [i.e., $I_f^T(\lambda)$] values computed excluding E_d015 and increasing the CHORS and JRC values by 0.5% to approximately account for the use of tap water instead of seawater are given in Table 6 together with their estimated maximum uncertainties and average $I_f(\lambda)$ values for the old [i.e., $I_f^O(\lambda)$] and new [i.e., $I_f^N(\lambda)$] subsets of radiometers. The maximum estimated uncertainties provided in Table 6 suggest the need for determining the $I_f(\lambda)$ values for each single diffuser when the intended use of the radiometer requires an absolute calibration at the 1% level (e.g., ocean color calibration and validation activities).

d. $I_f(\lambda)$ spectral dependence

Smith (1969), Petzold and Austin (1988), and Mueller (1995) observed an almost linear change of $I_f(\lambda)$ as a function of λ, with some deviation from linearity in the blue. Such a linear dependence is not observed in the current $I_f(\lambda)$ spectral measurements for the OCI-200 series of radiometers. The explanation for this is given by differences in the optical design of the radiometers involved in the various studies. The instruments analyzed by the aforementioned authors all had a single collector, while the OCI-200 series of radiometers has multiple collectors (one per spectral channel) designed to ensure optimum scattering and absorption performances at a specific center wavelength. This latter point is supported by the absorbance measurements made on sample OCI-200 collectors at six different center wavelengths (412, 443, 490, 555, 665, and 683 nm). The net absorbance (i.e., the difference between the absorbance measured in reflection mode subtracted from the absorbance measured in transmission mode) and the reflection–absorbance (i.e., the absorbance measured in reflection mode), obtained with a double-beam spectrometer equipped with an integrating sphere, are plotted in Fig. 5. For each collector, the net absorbance is quite linear as a function of wavelength, with some change in slope below 420 nm. The different collectors show very large relative differences appearing as an offset with some slight change in slope. Because of this, the spectral shape of the net absorbance resulting from values associated with each collector (i.e., the encircled numbers in Fig. 5) highlights a dependence that is not linear with wavelength.

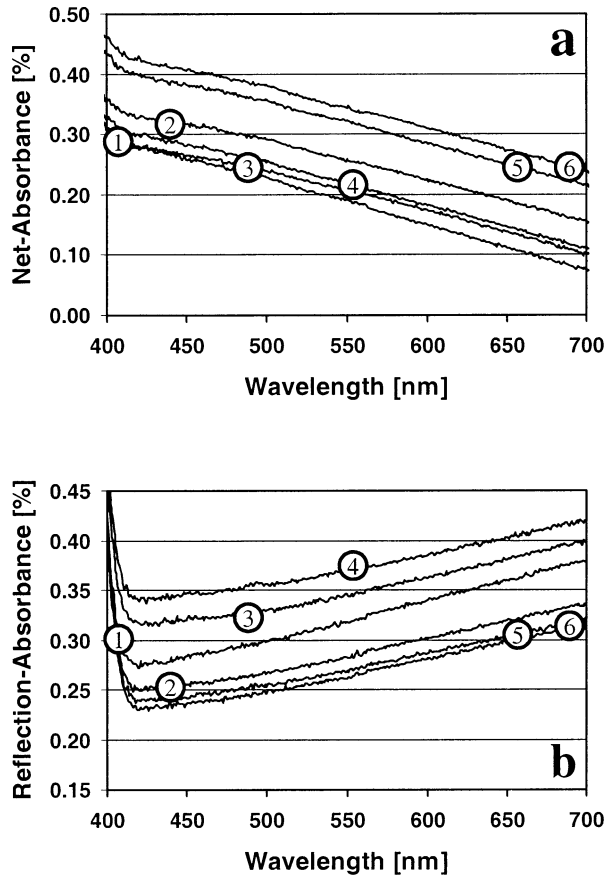

FIG. 5. The (a) net absorbance and (b) reflection–absorbance of a sample of OCI-200 collectors. The encircled numbers highlight the nominal working center wavelength (1, 412; 2, 443; 3, 490; 4, 555; 5, 665; and 6, 683 nm).

The comparison of Fig. 4 with Figs. 5a and 5b in general shows high values of spectral net absorbance

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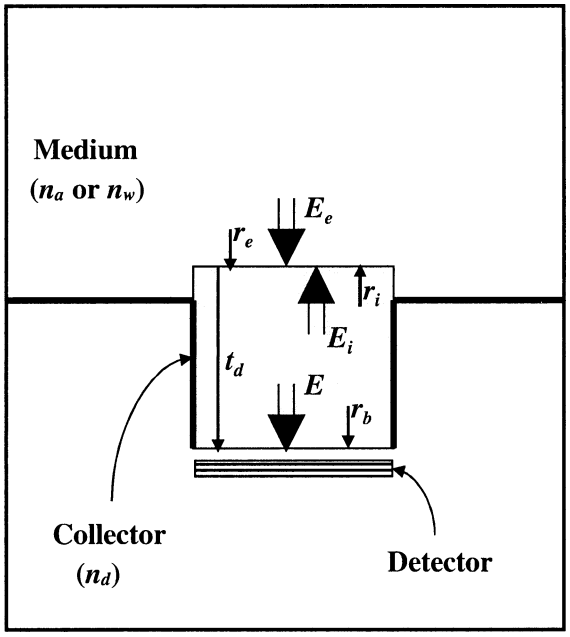

FIG. 6. Schematic of the reflectance and transmittance elements presented in Eq. (7).

and low values of spectral reflection–absorbance in correspondence to high $I_f(\lambda)$ values. Starting from this observation, a qualitative explanation of the $I_f(\lambda)$ spectral dependence can be constructed from the approximate equation describing the irradiance $E(\lambda)$ (Fig. 6) received by the detector located on the back of the collector:

$$E(\lambda) = E_e(\lambda)[1 - r_e(\lambda)]t_d(\lambda)[1 - r_b(\lambda)] + E_i(\lambda)r_i(\lambda), \quad (7)$$

where $E_e(\lambda)$ is the irradiance incident at the external surface of the collector, $r_e(\lambda)$ is the external reflectance of the medium–collector interface, $t_d(\lambda)$ is the collector transmittance that depends only on the diffuser material, $r_b(\lambda)$ is the reflectance of the collector–detector interface that does not depend on the refractive index of the external medium, and $E_i(\lambda)r_i(\lambda)$ identifies the diffuse irradiance reflected by the internal collector–medium interface with $r_i(\lambda)$ producing much higher effects than $r_e(\lambda)$ on $E(\lambda)$. When the medium in contact with the diffuser varies, and neglecting changes in $t_d(\lambda)$, $r_b(\lambda)$, and $r_e(\lambda)$, changes in $E_i(\lambda)r_i(\lambda)$ explain the variations in $I_f(\lambda)$ (i.e., when $E_i(\lambda)r_i(\lambda)$ increases, $E(\lambda)$ increases and $I_f(\lambda)$ decreases). Assuming as a first approximation that $E_i(\lambda)r_i(\lambda)$ is inversely related to the reflection–absorbance, the $I_f(\lambda)$ spectral dependence inversely follows the reflection–absorbance spectral dependence. The expected linear dependence with wavelength is very evident in $I_f(\lambda)$ values from in-water radiometers equipped with a single diffuser (Smith 1969; Mueller 1995). For the OCI-200 radiometers, the former spectral dependence can only be qualitative. In fact the difference in the reflection–absorbance among diffusers optimized for different center wavelengths results from slightly different optical characteristics of the different diffusers. The latter different optical characteristics may create different spectral dependences of $r_e(\lambda)$, $r_b(\lambda)$, $r_i(\lambda)$, and $t_d(\lambda)$ among diffusers and, consequently, induce slightly different spectral dependences of $I_f(\lambda)$ on the reflection–absorbance for each diffuser. This leads to the generic spectral dependence observed between the OCI-200 $I_f(\lambda)$ and the reflection–absorbance spectral values, but does not permit any quantitative comparison between the two. This finding suggests the inappropriateness, for the OCI-200 series of radiometers, of using the defined typical $I_f(\lambda)$ values to extrapolate data at wavelengths not included in the current analysis.

7. Conclusions

The present study, through multiple characterizations of the same radiometer, has demonstrated the possibility of achieving a measurement repeatability that is on average better than 0.6% (expressed by 2σ) on the characterization of $I_f(\lambda)$. Interlaboratory comparisons showed relative average uncertainties to be generally less than $\pm 0.6\%$ for both the reference radiometer and the multiple radiometer analyses (Tables 3 and 4). Similarly the method precision showed average values of 1.2% for both the single- (reference) and the multiple-radiometer analyses. The level of agreement between the single- and multiple-radiometer results establishes the consistency of the statistical approach (i.e., the number of characterizations was sufficient to quantify the primary analytical variables) and of the $I_f(\lambda)$ methodology (i.e., the major source of differences was associated with the water quality at each laboratory). The variability in $I_f(\lambda)$, within a sample of nine OCI-200 radiometers covering about 10% of the instrument production from 1994 to 1999, showed average dispersion values on the order of 2% with spectral values as high as 5%. An attempt at producing typical $I_f(\lambda)$ values for the OCI-200 series of radiometers showed maximum uncertainties spectrally varying from $\pm 1.4\%$ to $\pm 3.4\%$. This fully confirms the need, already pointed out by Mueller (1995), of ensuring a full spectral characterization of $I_f(\lambda)$values for each in-water radiometer required for an accurate absolute determination of irradiances [i.e., $E_u(z, \lambda)$, $E_d(z, \lambda)$] and derived quantities [i.e., $Q_n(z, \lambda)$, $R(z, \lambda)$].

An analysis of the sources of intra- and interlaboratory differences showed that an accuracy increase together with a standardization in $I_f(\lambda)$ characterization can be obtained by taking the following steps:

1) ensuring the best possible water purity to reduce scattering effects of particles (practical solutions may require reduction in the size of tanks to handle relatively small volumes of pure water);
2) skimming the surface to remove floating particles,

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which may change the surface transmittance [the alternative use of soap requires further investigation because soap slicks forming at the surface may spectrally affect the surface transmittance and thereby produce an overestimate of $I_f(\lambda)$];
3) using pure seawater instead of pure water (a practical alternative is to use pure water and apply a correction factor accounting for differences in the refractive indices between pure seawater and pure water);
4) applying quality assurance indices [like the $K(\lambda)$ values] to remove measurement sequences affected by a decreased quality of water or changes in the optical–mechanical setup;
5) using a small source and monitoring its stability during measurement sequences; and
6) minimizing the optical load of collectors by taking data at water depths larger than the radius of the area covered by the collectors.

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